

Methods — Team 14: Multimodal SDB Event Classification

1. Data

- **Source:** PATS (Pediatric Adenotonsillectomy Trial for Snoring) dataset, obtained via the National Sleep Research Resource (NSRR).
- **Format:** 494 MATLAB .mat files, one per subject (baseline home sleep apnea test).
- **Channels:** 7-channel Type II HSAT signals, sampled at ~20 Hz: thorax, airFlow, SpO2, arousalMean, aflowBreaths, sleepStages, OriginalDetections.
- **Labels:** Ground-truth apnea/hypopnea event start/end indices (events.apneas, events.hypopneas), plus subject-level AHI, apnea index, and hypopnea index.

Data quality exclusions (51 subjects removed → 443-subject clean cohort)

1. **Unverifiable sample-rate files (n = 13):** A subset of recordings had event indices generated assuming a fixed 20.0 Hz sample rate, but their actual measured rate deviated (16.0–21.33 Hz). Event timing was corrected by converting indices to seconds using a fixed nominal 20.0 Hz rather than each file's own measured rate. 108 of 121 affected files were recoverable this way (confirmed by restored overlap with OriginalDetections); 13 files had zero scored events and could not be verified either way, so they were excluded.
2. **Low signal-quality files (n = 38):** Files where the airFlow (nasal pressure) channel showed long stretches of flat or artifact-spiked signal inconsistent with normal breathing effort seen on thorax, causing the reference scoring to massively over- or under-detect apneas. Excluded rather than corrected, since the underlying channel data itself was unreliable.

2. Train/Validation Split

- **Level:** Subject-level (no window from the same subject appears in both train and validation).
- **Ratio:** 80% train / 20% validation.
- **Stratification:** By AHI severity bin (normal < 1, mild 1–5, moderate 5–10, severe > 10 events/hour) to ensure comparable severity distributions between splits, rather than a plain random split (which produced meaningfully different mean AHI between train and validation). The AHI 10 moderate/severe cutoff follows AASM-recommended **pediatric** criteria, appropriate here since PATS is a pediatric cohort (the adult standard uses AHI 15 for that cutoff).
- random_state = 42 for reproducibility.

3. Feature Engineering

- **Windowing:** Sliding window over each recording, 30-second window length, 15-second step (50% overlap).
- **Window label:** apnea or hypopnea if any portion of a scored event overlaps the window; otherwise normal.

- **Per-channel features (×7 channels):** mean, standard deviation, min, max, range, and linear slope — 42 features total per window.
- A shorter 10-second window was also tested and performed worse (macro F1 0.51 vs. 0.56 for 30 s), so 30 s/15 s was retained.

4. Model

- **Algorithm:** LightGBM multiclass gradient-boosted trees, classifying each window as normal, apnea, or hypopnea.
- **Class weighting:** Manually set weights {normal: 1, apnea: 5, hypopnea: 5} to counter severe class imbalance. `class_weight="balanced"` was tried first but overcorrected, driving precision down to ~0.25–0.27; the manual weights recovered precision to ~0.61 while keeping reasonable recall.

5. Event-Level Evaluation

Window-level predictions were converted into continuous events (consecutive same-label windows merged into one event) using a vectorized run-length approach, then scored against ground truth as follows:

- **Matching:** Overlap-based — a true event is padded by a ± 7.5 -second tolerance window, and any predicted event overlapping that padded window counts as a match. An earlier strict boundary-matching approach (requiring both start and end within tolerance) was far too harsh (Apnea F1 0.231, Hypopnea F1 0.342) and was abandoned.
- **Metrics:**
 - **Event-level Precision / Recall / F1**, computed separately for apnea and hypopnea.
 - **AHI MAE** — mean absolute error between the true AHI (events/hour) and the AHI computed from predicted events, averaged across validation subjects.

6. Threshold Tuning

- Class probabilities (`prob_apnea`, `prob_hypopnea`, `prob_normal`) from the model were used to search for better classification thresholds, independently per class and jointly (grid search over threshold pairs).
- Independent and joint tuning converged on essentially the same optimal thresholds, indicating the two classes' thresholds don't meaningfully interact for this problem.
- Threshold tuning traded off AHI accuracy for event F1: raising thresholds improved F1 but worsened AHI MAE (more on this in Results).

7. Results (Validation Set)

Metric	Baseline (0.5 threshold)	After threshold tuning
Apnea event F1	0.520 (P=0.487, R=0.559)	0.536
Hypopnea event F1	0.614	0.641
AHI MAE (events/hr)	1.84	2.06–2.10

Performance by AHI severity (baseline model):

Severity	n subjects	AHI MAE
Normal (AHI < 1)	34	0.19
Mild (1-5)	40	0.69
Moderate (5-10)	8	2.83
Severe (> 10)	7	18.56

8. Limitations

- Performance degrades sharply with severity: the model is highly accurate for normal/mild cases but substantially undercounts events in severe cases. A likely explanation is that in high-event-density recordings, back-to-back or closely spaced apnea/hypopnea events get merged into a single detected event by the window-to-event conversion step, undercounting the true event count even when the underlying windows are correctly classified.
- The airFlow channel's reliability issue (see exclusions above) suggests the remaining cohort may still contain some residual channel-quality variability not fully captured by the exclusion criteria.
- Threshold tuning is a precision/recall vs. AHI-accuracy tradeoff, not a free improvement — the tuned thresholds that raise F1 make AHI estimates slightly less accurate.